Advanced Econometrics

General-to-Specific

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General-to-Specific Approach (Gets)

Main idea:

- Start with a **general model**
- Include all variables that may be relevant
- Then simplify the model step by step

Goal:

- Keep the important variables
- Remove unnecessary ones
- Obtain a **parsimonious** final model

From a rich model to a simpler, interpretable model

Why Use General-to-Specific?

Why not start with a very small model?

- We may omit important variables
- Omitted variables can bias coefficient estimates
- A larger initial model is often safer

Gets logic:

- Start broad
- Test carefully
- Remove variables only when justified

Better to simplify a rich model than build up from a poor one

General-to-Specific: Basic Algorithm

1. Estimate the **general model**
2. Identify statistically insignificant variables
3. Test whether they are **jointly insignificant**
4. If yes, remove them together
5. If no, remove the **least significant** variable
6. Re-estimate the model
7. Repeat until the final model is satisfactory

Simplify gradually, not mechanically

General and Specific Model

General model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + u$$

Some coefficients may be statistically insignificant.

Specific model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_3 X_3 + \beta_5 X_5 + u$$

Question:

- Can we remove $X_2, X_4, \ldots$ without losing important information?

Why Joint Tests Matter

Suppose several variables look insignificant one by one.

Important:

- They may still be **jointly significant**
- So we should not remove them only because individual p-values are large

We test:

$$H_0 : \beta_2 = \beta_4 = \beta_6 = 0$$

Interpretation:

- If we do not reject H_0, these variables may be dropped
- If we reject H_0, at least one of them matters

Two Possible Cases

After estimating the general model:

Case 1: insignificant variables are jointly insignificant

- Remove them all at once

Case 2: insignificant variables are jointly significant

- Do not remove them all
- Remove variables one by one
- Usually start with the least significant variable

This is exactly what happens in your lab example

Exercise 1: Wage Equation

We start with a broad wage model including:

- age
- race
- marital status
- education
- region
- union membership
- hours worked
- experience and tenure
- industry dummies

Goal:

- Explain wage
- Remove variables that do not add useful explanatory power

Exercise 1: What Happens?

- First, estimate the **general model**
- Some regressors are insignificant
- Test whether all insignificant variables can be dropped jointly

Result:

- They are **jointly significant**
- So we cannot remove them all at once

Therefore:

- Remove the most insignificant variable
- Re-estimate the model
- Test again

Exercise 1: Step-by-Step Elimination

In the lab, variables are removed gradually, for example:

- trade
- personal services
- age
- agriculture
- mining
- professional services
- hours
- married
- race: other
- entertainment

At each step:

- re-estimate the model
- verify whether the removed set can be excluded

How the Joint Hypothesis Grows

At each step, the set of excluded variables becomes larger.

Examples:

$$H_0 : \beta_{\text{trade}} = 0$$

then

$$H_0 : \beta_{\text{trade}} = \beta_{\text{perserv}} = 0$$

then

$$H_0 : \beta_{\text{trade}} = \beta_{\text{perserv}} = \beta_{\text{age}} = 0$$

Idea:

- Every newly removed variable is added to the restriction set

When Do We Stop?

We stop when:

- the remaining regressors are statistically significant
- the excluded variables can be omitted
- the model is simpler and still economically sensible

Final result:

- a smaller model
- easier interpretation
- no unnecessary regressors

This is the final specific model

Exercise 3: Gets in a Panel Model

The same logic can be used in panel data.

In the lab:

- the model is estimated with **fixed effects**
- some coefficients are insignificant
- joint restrictions are tested with a **Wald test**

So Gets is not only for simple OLS:

- it can also be applied in panel settings

OLS Version vs Panel Version

OLS case:

- Estimate with `lm()`
- Test restrictions with `anova()` or `linearHypothesis()`

Panel fixed-effects case:

- Estimate with `plm(..., model="within")`
- Test restrictions with a **Wald test**

Same idea in both cases:

start general $\rightarrow$ test $\rightarrow$ simplify

Advantages of General-to-Specific

- Starts from a rich specification
- Reduces risk of omitted variable bias at the beginning
- Produces a more parsimonious final model
- Encourages formal testing during simplification

Useful when:

- theory suggests many possible regressors
- we want a data-supported simplification

Limitations and Cautions

General-to-specific is useful, but:

- Results may depend on the initial general model
- Repeated testing may create data-mining concerns
- Statistical significance is not everything
- Economic theory should still guide decisions

A good final model should be statistically and economically sensible

Summary: General-to-Specific

- Begin with a general model
- Test insignificant variables jointly
- If possible, remove them together
- Otherwise, eliminate variables one by one
- Re-estimate and test after each step
- Stop when the model is parsimonious and adequate

Gets = structured simplification of an econometric model

ANOVA vs Wald Test

Goal:

- Test whether some variables can be removed

ANOVA test:

- Compares two models (full vs reduced)

Wald test:

- Tests restrictions in one model
- Example: $\beta_1 = \beta_2 = \beta_3 = 0$

Interpretation:

- High p-value $\rightarrow$ variables can be removed
- Low p-value $\rightarrow$ keep them

Thank you